

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		• Hadrons are made up of quarks • Quarks / hadrons feel the strong force. • Hadrons can be split into two groups, baryons and mesons [Accept three groups and reference to anti-baryons] • Baryons contain three quarks. Protons and neutrons are baryons. • Protons contain 2 up quarks and 1 down quark. • Neutrons contain 1 up quark and 2 down quarks. • Up quarks carry a charge of $+\frac{2}{3}$, down quarks carry a charge of $-\frac{1}{3}$. • For protons: $+\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$ For neutrons: $+\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$. • Mesons contain a quark and an antiquark or detail e.g π^+ is $u\bar{d}$ ($+\frac{2}{3} + \frac{1}{3}$) etc • There are 3 generations of hadrons / quarks. • Hadrons have a lepton number of 0, baryons have a baryon number of 1, anti-baryons -1 and mesons 0. 5-6 marks A comprehensive analysis of baryons and mesons. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks A comprehensive account of either baryons or mesons or a limited account of both. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i>	6			6		

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			1-2 marks A limited account of either baryons or mesons. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)		Charge: $+1 -1 = 0$ $+1 -1$ (1) Baryon: $1 + 0 = 1 + 0 + 0$ (1) Accept: $uud + \bar{u}\bar{d} = udd + u\bar{d} + \bar{u}\bar{d}$ Lepton: $0 + 0 = 0 + 0 + 0$ (1) N.B. Charge and lepton number must be present and either baryon or quark number		3		3		
			Question 5 total	6	3	0	9	0	0